

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		15481	
Total Threats Mitigated	47	Total Packets Analyzed		4096486076	
High Priority	11	Malicious Packets		4083721559	
Low Priority	36	True Negatives Passed		10205	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		6	
ML Metrics	Value	SYN Flood		14	
False Positives	23	UDP Flood		11	
FP Rate	0.23%	Uncertain		16	
IF Precision	99.53%				
IF Recall	96.79%	Action		Count	
IF F1-Score	98.14%	Quarantined		10	
IF Accuracy	98.78%	Time Ban		10	
		Blackhole		0	
		Released		9	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.53%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	96.79%	Share of actual attacks successfully flagged
F1-Score	98.14%	Balanced measure combining Precision and Recall
Accuracy	98.78%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.23%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	3.21%	Actual attacks that went undetected
True Positive Rate (TPR)	96.79%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.77%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4829	FN 160
Actual: Normal	FP 23	TN 10045

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	99.98%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	99.99%	Balanced measure combining Precision and Recall
Accuracy	99.98%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	4.35%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	95.65%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	2491	0	0
Act: ICMP	0	831	0
Act: UDP	0	0	1507

2c. Hybrid Cascaded

Metric	Value	Description
Precision	99.53%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	96.79%	Share of actual attacks successfully flagged
F1-Score	98.14%	Balanced measure combining Precision and Recall
Accuracy	98.78%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.23%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	3.21%	Actual attacks that went undetected
True Positive Rate (TPR)	96.79%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.77%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4829	FN 160
Actual: Normal	FP 23	TN 10045

2d. Response Latency

Metric	Value	Description
Detection Time	910.1 ms	Average time taken to detect an attack
Mitigation Response Time	30.1 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.52%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	28.95%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	19.55%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.16	4	3	2026-09-25 14:44:59	2026-09-25 14:47:39
10.0.0.17	4	3	2026-09-25 14:45:00	2026-09-25 14:47:42
10.0.0.21	4	3	2026-09-25 14:45:29	2026-09-25 14:47:45
10.0.0.18	3	3	2026-09-25 14:45:00	2026-09-25 14:45:52
10.0.0.19	3	3	2026-09-25 14:45:05	2026-09-25 14:46:16
10.0.0.20	3	2	2026-09-25 14:45:30	2026-09-25 14:47:39
10.0.0.22	3	2	2026-09-25 14:45:30	2026-09-25 14:47:51
10.0.0.23	2	1	2026-09-25 14:46:00	2026-09-25 14:47:41

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.24	2	1	2026-09-25 14:46:01	2026-09-25 14:47:42
10.0.0.25	2	1	2026-09-25 14:46:02	2026-09-25 14:47:45
10.0.0.8	1	1	2026-09-25 14:43:17	2026-09-25 14:43:47
10.0.0.3	1	1	2026-09-25 14:43:40	2026-09-25 14:43:50

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 14:43:17	10.0.0.8	DDoS	UDP Flood	92.4%	High	Time Ban (30s)
2026-09-25 14:43:47	10.0.0.8	DDoS	UDP Flood	92.4%	High	Released
2026-09-25 14:43:50	10.0.0.3	DDoS	Uncertain	59.0%	Low	Released
2026-09-25 14:44:59	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:45:00	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:45:00	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:45:05	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:45:09	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Time Ban (30s)
2026-09-25 14:45:09	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Time Ban
2026-09-25 14:45:10	10.0.0.18	DDoS	SYN Flood	90.9%	Critical	Time Ban (30s)
2026-09-25 14:45:10	10.0.0.17	DDoS	SYN Flood	94.2%	Critical	Time Ban (30s)
2026-09-25 14:45:11	10.0.0.18	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-25 14:45:11	10.0.0.17	DDoS	SYN Flood	94.2%	Critical	Time Ban
2026-09-25 14:45:16	10.0.0.19	DDoS	SYN Flood	93.0%	Critical	Time Ban (30s)
2026-09-25 14:45:16	10.0.0.19	DDoS	SYN Flood	93.0%	Critical	Time Ban
2026-09-25 14:45:29	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:45:30	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:45:30	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:45:40	10.0.0.21	DDoS	UDP Flood	84.3%	High	Time Ban (30s)
2026-09-25 14:45:40	10.0.0.21	DDoS	UDP Flood	84.3%	High	Time Ban
2026-09-25 14:45:41	10.0.0.18	DDoS	SYN Flood	90.9%	High	Released
2026-09-25 14:45:41	10.0.0.20	DDoS	UDP Flood	86.3%	Critical	Time Ban (30s)
2026-09-25 14:45:41	10.0.0.22	DDoS	UDP Flood	86.5%	Critical	Time Ban (30s)
2026-09-25 14:45:41	10.0.0.22	DDoS	UDP Flood	86.5%	Critical	Time Ban
2026-09-25 14:45:41	10.0.0.20	DDoS	UDP Flood	86.3%	Critical	Time Ban
2026-09-25 14:45:52	10.0.0.18	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-25 14:46:00	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:46:01	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:46:02	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:46:10	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Time Ban (1m)
2026-09-25 14:46:10	10.0.0.21	DDoS	UDP Flood	84.3%	High	Released
2026-09-25 14:46:10	10.0.0.23	DDoS	ICMP Flood	90.8%	High	Time Ban (30s)
2026-09-25 14:46:10	10.0.0.23	DDoS	ICMP Flood	90.8%	Critical	Time Ban
2026-09-25 14:46:11	10.0.0.17	DDoS	SYN Flood	94.6%	Critical	Time Ban (1m)
2026-09-25 14:46:11	10.0.0.24	DDoS	ICMP Flood	91.2%	High	Time Ban (30s)
2026-09-25 14:46:11	10.0.0.24	DDoS	ICMP Flood	91.2%	High	Time Ban
2026-09-25 14:46:13	10.0.0.25	DDoS	ICMP Flood	91.2%	High	Time Ban (30s)

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 14:46:13	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Time Ban
2026-09-25 14:46:16	10.0.0.19	DDoS	SYN Flood	93.0%	Critical	Time Ban (1m)
2026-09-25 14:46:20	10.0.0.21	DDoS	UDP Flood	84.3%	High	Time Ban (1m)
2026-09-25 14:47:39	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (2m)
2026-09-25 14:47:39	10.0.0.20	DDoS	UDP Flood	82.3%	Critical	Time Ban (1m)
2026-09-25 14:47:41	10.0.0.23	DDoS	Uncertain	-	Low	Released
2026-09-25 14:47:42	10.0.0.17	DDoS	Uncertain	-	Low	Released
2026-09-25 14:47:42	10.0.0.24	DDoS	Uncertain	-	Low	Released
2026-09-25 14:47:45	10.0.0.25	DDoS	Uncertain	-	Low	Released
2026-09-25 14:47:51	10.0.0.22	DDoS	Uncertain	-	Low	Released

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.22	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-25 14:47:51	Attack Stopped
10.0.0.21	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-25 14:47:45	Attack Stopped
10.0.0.25	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-25 14:47:45	Attack Stopped
10.0.0.24	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-25 14:47:42	Attack Stopped
10.0.0.17	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-25 14:47:42	Attack Stopped
10.0.0.23	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-25 14:47:41	Attack Stopped
10.0.0.20	UDP Flood	0.6443	82.3%	Critical	Phase 2	1	10s	2026-09-25 14:47:39	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6476	92.9%	Critical	Phase 2	1	11s	2026-09-25 14:47:39	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6276	84.3%	High	Phase 2	3	10s	2026-09-25 14:46:20	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6384	93.0%	Critical	Phase 2	3	1m 11s	2026-09-25 14:46:16	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6454	91.2%	High	Phase 2	1	11s	2026-09-25 14:46:13	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6446	91.2%	High	Phase 2	1	10s	2026-09-25 14:46:11	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6318	86.9%	High	Phase 2	2	41s	2026-09-25 14:46:11	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.20	UDP Flood	0.6316	87.0%	High	Phase 2	2	41s	2026-09-25 14:46:11	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6341	94.6%	Critical	Phase 2	3	1m 11s	2026-09-25 14:46:11	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6440	90.8%	High	Phase 2	1	10s	2026-09-25 14:46:10	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6293	84.3%	High	Phase 2	1	41s	2026-09-25 14:46:10	Ban Expired
10.0.0.16	SYN Flood	0.6276	93.3%	Critical	Phase 2	3	1m 11s	2026-09-25 14:46:10	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6322	90.9%	Critical	Phase 2	3	11s	2026-09-25 14:45:52	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6481	93.0%	Critical	Phase 2	2	41s	2026-09-25 14:45:46	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6398	86.5%	Critical	Phase 2	1	11s	2026-09-25 14:45:41	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	86.3%	Critical	Phase 2	1	11s	2026-09-25 14:45:41	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.5944	94.6%	High	Phase 2	2	41s	2026-09-25 14:45:41	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6012	90.9%	High	Phase 2	1	41s	2026-09-25 14:45:41	Ban Expired
10.0.0.21	UDP Flood	0.6300	84.3%	High	Phase 2	1	11s	2026-09-25 14:45:40	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6328	93.3%	Critical	Phase 2	2	41s	2026-09-25 14:45:40	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6518	93.0%	Critical	Phase 2	1	11s	2026-09-25 14:45:16	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6229	94.2%	Critical	Phase 2	1	10s	2026-09-25 14:45:10	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6392	90.9%	Critical	Phase 2	1	10s	2026-09-25 14:45:10	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6380	93.3%	Critical	Phase 2	1	10s	2026-09-25 14:45:09	Attack persisted - escalating to time ban
10.0.0.3	Uncertain	0.6045	59.0%	Low	Phase 1	1	10s	2026-09-25 14:43:50	Attack Stopped
10.0.0.8	UDP Flood	0.5776	92.4%	High	Phase 2	1	30s	2026-09-25 14:43:47	Ban Expired

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME
Network Administrator

DATE